

Writing Assignment 1

Instructions. Choose any 5 of the 15 problems to complete. You may either write your solutions by hand and submit scanned copies or clear photos, or type your solutions using LaTeX or Microsoft Word. All submissions must be electronic; paper submissions will not be accepted. Writing Assignment 1 is worth 10 points toward your total grade.

Conventions. All logarithms are natural. Unless stated otherwise, training and test examples follow the same distribution, samples are i.i.d., and binary labels lie in $\{0, 1\}$. For a joint distribution P on $\mathcal{X} \times \mathcal{Y}$, write

$$R(h) = \mathbb{P}_{(X,Y) \sim P}(h(X) \neq Y), \quad \hat{R}_S(h) = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{h(X_i) \neq Y_i\}.$$

In realizable problems, $Y = c(X)$ with $c \in \mathcal{H}$. A distribution on inputs alone is denoted by D . When a problem uses sign-valued predictions, labels lie in $\{-1, +1\}$ instead. Classes are nonempty; assume the measurability needed for all displayed probabilities and suprema. ERM is treated as an oracle. Unless a result is the object of the problem, previously established results may be used. Round numerical sample sizes upward to integers.

Problem 1. Reading the PAC guarantee [MRT18, Sec. 2.1]

- (a) State realizable PAC learnability with the quantifiers over the learner, target concept, input distribution, accuracy ε , confidence δ , and sample size in the correct order. Identify the randomness in the probability statement.
- (b) Explain the difference between realizable and agnostic learning, and between proper and improper learning. Does a statistical PAC guarantee alone imply computational efficiency?
- (c) Let $\mathcal{X} = \{a, b\}$, $c(a) = 0$, and $c(b) = 1$. Training inputs are always a , while test inputs are always b . Compute the training and test risks of $h(x) \equiv 0$. Explain precisely which assumption of the usual generalization guarantee fails.

Problem 2. Finite classes and consistent learners [MRT18, Sec. 2.2]

Let $|\mathcal{H}| = N < \infty$, assume realizability, and let $h_S \in \mathcal{H}$ be any consistent output.

- (a) For a fixed h with $R(h) > \varepsilon$, bound the probability that $\hat{R}_S(h) = 0$. Use a union bound to prove

$$\mathbb{P}(R(h_S) > \varepsilon) \leq Ne^{-m\varepsilon}.$$

- (b) Derive a sufficient sample size for accuracy ε and confidence $1 - \delta$. Evaluate it when $N = 1001$ and $\varepsilon = \delta = 0.05$.
- (c) For conjunctions over n Boolean variables, each variable is included positively, negatively, or omitted. Count the class, and evaluate the bound for $n = 20$, $\varepsilon = 0.10$, and $\delta = 0.05$.
- (d) Quantify how the sufficient sample size changes when N doubles, when ε is halved, and when δ is halved.

Problem 3. Hoeffding's inequality from Taylor's theorem

Let $Z_1, \dots, Z_m$ be independent, with $Z_i \in [0, 1]$ and $\mu_i = \mathbb{E}Z_i$. Put $W_i = Z_i - \mu_i$ and $\psi_i(\lambda) = \log \mathbb{E}e^{\lambda W_i}$. For each i , define

$$\mathbb{E}_\lambda[f(Z_i)] = \frac{\mathbb{E}[f(Z_i)e^{\lambda W_i}]}{\mathbb{E}e^{\lambda W_i}}.$$

- (a) Differentiate ψ_i to show that $\psi_i(0) = \psi'_i(0) = 0$ and $\psi''_i(\lambda) = \text{Var}_\lambda(Z_i)$. Explain why the tilted law still has support in $[0, 1]$.

- (b) Use

$$\text{Var}_\lambda(Z_i) \leq \mathbb{E}_\lambda(Z_i - \frac{1}{2})^2 \leq \frac{1}{4}$$

and Taylor's theorem to prove $\mathbb{E}e^{\lambda W_i} \leq e^{\lambda^2/8}$.

- (c) Apply Markov's inequality and independence, optimize over $\lambda > 0$, and obtain

$$\mathbb{P}\left(\left|\frac{1}{m} \sum_{i=1}^m (Z_i - \mu_i)\right| \geq t\right) \leq 2e^{-2mt^2}, \quad t > 0.$$

Problem 4. Uniform convergence and approximate ERM [MRT18, Sec. 2.3]

Let $\mathcal{H}$ be finite, and allow arbitrary stochastic labels. An approximate ERM output $\hat{h} \in \mathcal{H}$ satisfies

$$\hat{R}_S(\hat{h}) \leq \inf_{h \in \mathcal{H}} \hat{R}_S(h) + \eta, \quad \eta \geq 0.$$

- (a) Prove that, with probability at least $1 - \delta$,

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_S(h)| \leq \gamma_m := \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2m}}.$$

- (b) Show that on this event, $R(\hat{h}) - \inf_{h \in \mathcal{H}} R(h) \leq 2\gamma_m + \eta$. Label the step that uses approximate ERM.
- (c) For $\eta = 0$, solve for a sufficient m giving excess risk at most ε . Compare its dependence on ε with Problem 2. For $\eta > 0$, what restriction on η is needed for this argument to certify excess risk at most ε ?

Problem 5. Why reusing training data changes the argument [MRT18, Secs. 2.2–2.3]

Let D be uniform on $\mathcal{X} = \{1, \dots, N\}$, where $N > m$, and suppose all labels equal zero. The class contains every binary function on $\mathcal{X}$. Define $h_S(x) = 0$ on inputs that occur in the training sample and $h_S(x) = 1$ elsewhere.

- (a) Show that $\hat{R}_S(h_S) = 0$ and $R(h_S) \geq 1 - m/N$ for every sample. Why is this output an ERM solution?
- (b) Explain why applying fixed-hypothesis Hoeffding directly to h_S would be invalid.
- (c) Let V be a separate validation sample of size r , independent of S , and use it only to evaluate h_S . Condition on S and prove

$$\mathbb{P}_{S,V}\left(|R(h_S) - \hat{R}_V(h_S)| > \sqrt{\frac{\log(2/\delta)}{2r}}\right) \leq \delta.$$

Explain why this bound has no $\log |\mathcal{H}|$ term, and why the same justification can fail if V is used to choose among predictors.

Problem 6. Computing complexity on small samples [MRT18, Sec. 3.1]

For a fixed sample $S = (z_1, \dots, z_m)$, use

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_\sigma \sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i), \quad \mathbb{P}(\sigma_i = 1) = \mathbb{P}(\sigma_i = -1) = \frac{1}{2}.$$

- Compute the complexity of a singleton $\{g_0\}$. Explain why the magnitude or variation of g_0 does not change this answer.
- For $m = 4$ and the two constant functions $g \equiv -1$ and $g \equiv +1$, compute the complexity exactly by grouping sign vectors according to their number of positive entries.
- Let $\mathcal{X} = \{a, b, c, d\}$ and let $\mathcal{H}$ contain all maps $\mathcal{X} \rightarrow \{-1, +1\}$. Compute $\hat{\mathfrak{R}}_S(\mathcal{H})$ for $S = (a, a, b, b)$ and for $S = (a, b, c, d)$. Explain the effect of repeated inputs.
- If X_i are i.i.d. uniform on $\{a, b\}$ and $m = 2$, compute the expected complexity $\mathfrak{R}_2(\mathcal{H})$ for the class of all sign-valued maps on $\{a, b\}$.

Problem 7. Properties and convex combinations [MRT18, Sec. 3.1]

Fix a sample S . All classes below are uniformly bounded on S .

- Prove monotonicity under class inclusion, positive homogeneity $\hat{\mathfrak{R}}_S(c\mathcal{G}) = c\hat{\mathfrak{R}}_S(\mathcal{G})$ for $c \geq 0$, and translation invariance $\hat{\mathfrak{R}}_S(\mathcal{G} + a) = \hat{\mathfrak{R}}_S(\mathcal{G})$ for a fixed function a .
- For $\mathcal{G}_1 + \mathcal{G}_2 = \{g_1 + g_2 : g_j \in \mathcal{G}_j\}$, prove subadditivity of empirical Rademacher complexity. Can the bound be sharpened to equality when g_1 and g_2 are chosen independently from their respective classes?
- Let $\text{conv}(\mathcal{G})$ denote all finite convex combinations of functions in $\mathcal{G}$. Prove

$$\hat{\mathfrak{R}}_S(\text{conv}(\mathcal{G})) = \hat{\mathfrak{R}}_S(\mathcal{G}).$$

- Explain why the previous identity does not directly apply to the class obtained by thresholding those convex combinations to produce majority-vote classifiers.

Problem 8. From prediction complexity to loss complexity [MRT18, Sec. 3.1]

Use sign labels $Y \in \{-1, +1\}$ and a class $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$. Let

$$\mathcal{G} = \{g_h : (x, y) \mapsto \mathbf{1}\{h(x) \neq y\} : h \in \mathcal{H}\}.$$

- Prove that for a labeled sample S and its input sample S_X ,

$$\hat{\mathfrak{R}}_S(\mathcal{G}) = \frac{1}{2} \hat{\mathfrak{R}}_{S_X}(\mathcal{H}).$$

Use $\mathbf{1}\{h(x) \neq y\} = (1 - yh(x))/2$ and explain why multiplying each Rademacher sign by a fixed label preserves its distribution.

- Deduce the corresponding identity between expected complexities, with the loss class averaged over the joint distribution and the hypothesis class over the input marginal.
- Given the bounded-loss theorem

$$\mathbb{E}g \leq \frac{1}{m} \sum_i g(Z_i) + 2\mathfrak{R}_m(\mathcal{G}) + \sqrt{\frac{\log(1/\delta)}{2m}},$$

which holds uniformly with probability $1 - \delta$, derive the classification bound. Why is the coefficient of $\mathfrak{R}_m(\mathcal{H})$ equal to one?

Problem 9. A guided symmetrization proof [MRT18, Sec. 3.1]

Let $\mathcal{G} \subseteq [0, 1]^{\mathcal{Z}}$, $S = (Z_1, \dots, Z_m) \sim P^m$, and

$$\Phi(S) = \sup_{g \in \mathcal{G}} \left(\mathbb{E}_P g - \frac{1}{m} \sum_i g(Z_i) \right).$$

You may use McDiarmid's inequality: if changing coordinate i changes F by at most c_i , then $\mathbb{P}(F - \mathbb{E}F > t) \leq \exp(-2t^2 / \sum_i c_i^2)$.

- (a) Show that changing one point changes Φ by at most $1/m$, and obtain a high-probability bound on $\Phi(S) - \mathbb{E}\Phi(S)$.
- (b) Introduce an independent ghost sample S' . Justify

$$\mathbb{E}\Phi(S) \leq \mathbb{E}_{S, S'} \sup_g \frac{1}{m} \sum_i (g(Z'_i) - g(Z_i)).$$

- (c) Use pairwise exchangeability and independent signs to show that the right side is at most $2\mathfrak{R}_m(\mathcal{G})$. Explain why the supremum generally cannot be moved outside the sign expectation.
- (d) Prove that $\hat{\mathfrak{R}}_S(\mathcal{G})$ also has bounded differences $1/m$. Apply McDiarmid to its lower tail and combine two events of failure probability $\delta/2$ to obtain

$$\mathbb{E}_P g \leq \frac{1}{m} \sum_i g(Z_i) + 2\hat{\mathfrak{R}}_S(\mathcal{G}) + 3\sqrt{\frac{\log(2/\delta)}{2m}}$$

uniformly over $g \in \mathcal{G}$.

Problem 10. Massart's lemma and evaluation vectors [MRT18, Sec. 3.2]

Let $A \subset \mathbb{R}^m$ be finite and nonempty, and let $r = \max_{a \in A} \|a\|_2$. You may use $\mathbb{E}e^{t\sigma} = \cosh(t) \leq e^{t^2/2}$ for a Rademacher sign σ .

- (a) Using Jensen's inequality and a sum over A , prove that for every $\lambda > 0$,

$$\mathbb{E}_\sigma \max_{a \in A} \langle \sigma, a \rangle \leq \frac{\log |A|}{\lambda} + \frac{\lambda r^2}{2}.$$

- (b) Optimize over λ and handle $r = 0$ and $|A| = 1$ separately to obtain Massart's lemma.
- (c) For a sign-valued class $\mathcal{H}$, let

$$N_{S_X} = |\{(h(x_1), \dots, h(x_m)) : h \in \mathcal{H}\}|, \quad \Pi_{\mathcal{H}}(m) = \max_{x_1, \dots, x_m} N_{S_X}.$$

Apply the lemma to these evaluation vectors to show

$$\hat{\mathfrak{R}}_{S_X}(\mathcal{H}) \leq \sqrt{\frac{2 \log N_{S_X}}{m}} \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}}.$$

Explain why one counts evaluation vectors rather than parameter settings, and why any such complexity is also at most one.

Problem 11. Exact growth functions on the line [MRT18, Secs. 3.2–3.3]

Consider increasing thresholds $h_t(x) = \mathbf{1}\{x \geq t\}$ and intervals $h_{a,b}(x) = \mathbf{1}\{a \leq x \leq b\}$ on $\mathbb{R}$.

- (a) On m distinct ordered points, count their labelings and prove

$$\Pi_{\text{thr}}(m) = m + 1, \quad \Pi_{\text{int}}(m) = 1 + \frac{m(m+1)}{2}.$$

- (b) List the interval labelings on three ordered points. Which one is missing? Use constructive lower bounds and universal upper bounds to determine the VC dimension of both classes.
- (c) Bound the empirical Rademacher complexity of the sign-encoded versions of these classes on m points. Evaluate both bounds for $m = 100$, and state whether either exceeds the trivial upper bound one.

Problem 12. Axis-aligned boxes [MRT18, Sec. 3.3]

Let $\mathcal{H}_{\text{box}}$ contain indicators of closed axis-aligned boxes in $\mathbb{R}^p$, where $p \geq 1$.

- (a) Prove that the $2p$ points $\{e_1, -e_1, \dots, e_p, -e_p\}$ are shattered. Give an explicit way to choose each coordinate interval for an arbitrary subset of positive points, including the all-negative labeling.
- (b) For any set of $2p + 1$ distinct points, select at most $2p$ points attaining the coordinate minima and maxima. Construct a labeling that no box realizes. Explain why ties among extrema do not invalidate the argument.
- (c) Conclude that $\text{VCdim}(\mathcal{H}_{\text{box}}) = 2p$. Why would showing only one unshatterable configuration of $2p + 1$ points be insufficient?

Problem 13. Affine halfspaces and affine dependence [MRT18, Sec. 3.3]

Let $\mathcal{H}_{\text{aff}} = \{x \mapsto \mathbf{1}\{w^\top x + b \geq 0\} : w \in \mathbb{R}^p, b \in \mathbb{R}\}$.

- (a) For $p + 1$ affinely independent points, use the augmented vectors $(x_i, 1)$ to solve for scores $w^\top x_i + b \in \{-1, +1\}$ matching any prescribed labeling. Deduce a VC lower bound.
- (b) Any $p + 2$ points admit coefficients α_i , not all zero, with $\sum_i \alpha_i = 0$ and $\sum_i \alpha_i x_i = 0$. Label points with $\alpha_i > 0$ positive and those with $\alpha_i < 0$ negative; assign arbitrary labels when $\alpha_i = 0$. Prove that no affine halfspace realizes this labeling. Account for the convention that zero score is positive.
- (c) Conclude $\text{VCdim}(\mathcal{H}_{\text{aff}}) = p + 1$. Explain why merely counting $p + 1$ real parameters would not constitute a proof.

Problem 14. Sauer's lemma and the growth dichotomy [MRT18, Sec. 3.3]

Let $G \subseteq \{0, 1\}^m$ have VC dimension at most d . Removing its last coordinate gives a prefix family G_1 ; let G_2 contain the prefixes occurring with both final labels.

- (a) Prove $|G| = |G_1| + |G_2|$, $\text{VCdim}(G_1) \leq d$, and $\text{VCdim}(G_2) \leq d - 1$ when $G_2 \neq \emptyset$ and $d \geq 1$.
- (b) Use induction and Pascal's identity to prove

$$\Pi_{\mathcal{H}}(m) \leq \sum_{j=0}^{\min(d,m)} \binom{m}{j}$$

whenever $\text{VCdim}(\mathcal{H}) \leq d$. State the boundary cases, including an empty family of prefixes.

- (c) For $m \geq d \geq 1$, put $u = d/m \in (0, 1]$ and show

$$u^d \sum_{j=0}^d \binom{m}{j} \leq \sum_{j=0}^d \binom{m}{j} u^j \leq (1+u)^m \leq e^{mu}.$$

Deduce $\Pi_{\mathcal{H}}(m) \leq (em/d)^d$. Compare the resulting binomial bound with the exact interval growth function in Problem 11.

- (d) Explain why finite VC dimension implies polynomial worst-case growth, whereas infinite VC dimension gives $\Pi_{\mathcal{H}}(m) = 2^m$ for every m .

Problem 15. From VC dimension to a data budget [MRT18, Secs. 3.1–3.3]

Let $\mathcal{H}$ have VC dimension d , with $m \geq d \geq 1$. Use sign encoding when applying Rademacher bounds.

- (a) Combine Problems 8, 10, and 14 to prove that, with probability $1 - \delta$, uniformly over h ,

$$R(h) \leq \hat{R}_S(h) + B_1(m, d, \delta), \quad B_1(m, d, \delta) = \sqrt{\frac{2d \log(em/d)}{m}} + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

- (b) Apply the bounded-loss theorem also to complementary losses and allocate failure probability $\delta/2$ to each direction. Prove

$$\sup_h |R(h) - \hat{R}_S(h)| \leq B_2(m, d, \delta), \quad B_2(m, d, \delta) = \sqrt{\frac{2d \log(em/d)}{m}} + \sqrt{\frac{\log(2/\delta)}{2m}}.$$

Why does the one-sided bound alone not establish this two-sided statement?

- (c) Deduce an excess-risk bound $2B_2$ for an exact ERM output $\hat{h} \in \mathcal{H}$, using an approximating sequence if the population infimum is not attained.
- (d) For $m = 5000$ and $\delta = 0.05$, compute B_1 and B_2 for intervals ($d = 2$) and affine halfspaces in $\mathbb{R}^{20}$ ($d = 21$). For intervals, find the smallest m among 1000, 5000, 6000, 10000 for which $B_1 \leq 0.10$.

References

- [MRT18] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. *Foundations of Machine Learning*, second edition. MIT Press, 2018. Chapter 2: The PAC Learning Framework; Chapter 3: Rademacher Complexity and VC-Dimension. [Author's book page and official online editions](#).